

Analysis de Regressión

Función normal

$$f(y|\mu) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]$$

$$\log -2 \log(y-\mu) - 2 \log(2\sigma) = -2(\log(y-\mu) - \log(2\sigma))$$

~~$$\frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]$$~~
$$\log\left(\frac{1}{(2\pi\sigma^2)^{1/2}}\right) - \frac{(y-\mu)^2}{2\sigma^2}$$

$$\log 1 - \frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2 - 2ym + \mu^2}{2\sigma^2} = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2 - 2ym + \mu^2}{2\sigma^2}$$

$$= \frac{-2ym}{2\sigma^2} - \frac{y^2}{2\sigma^2} + \frac{\mu^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)$$

$$= \underbrace{\frac{-ym}{\sigma^2}}_{a(y)} + \underbrace{\frac{-y^2}{2\sigma^2}}_{b(y)} + \underbrace{\frac{\mu^2}{2\sigma^2}}_{c(\mu)} - \underbrace{\frac{1}{2} \log(2\pi\sigma^2)}_{d(y)}$$

Analysis de Regressión

Función normal

$$f(y|m) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left[-\frac{(y-m)^2}{2\sigma^2}\right]$$

$$\log f(y|m) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y-m)^2}{2\sigma^2}$$

~~$$\frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left[-\frac{(y-m)^2}{2\sigma^2}\right]$$~~

$$\log\left(\frac{1}{(2\pi\sigma^2)^{1/2}}\right) - \frac{(y-m)^2}{2\sigma^2}$$

$$\log f = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2 - 2ym + m^2}{2\sigma^2} = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2 - 2ym + m^2}{2\sigma^2}$$

$$= \frac{-2ym}{2\sigma^2} - \frac{y^2}{2\sigma^2} + \frac{m^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)$$

$$= \frac{-ym}{\sigma^2} - \frac{y^2}{2\sigma^2} + \frac{m^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)$$

$$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow \\ a(y) & b(y) & c(m) & d(y) \end{matrix}$$